

Clustering

K-Means Clustering Algorithm

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- 4 K-Means Properties
- 5 Challenges and Considerations
- 6 Choosing K — The Elbow Method
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What is Clustering?

Definition

An **unsupervised** learning technique — groups similar data points together without any labels.

Where is it used?

- Grouping emails or search results
- Customer shopping pattern analysis
- Segmenting regions of images

Useful when you don't know what patterns exist in your data.

Use With Care

Can produce **meaningless results** if the wrong distance metric or K is chosen.



Measuring Similarity

One option: squared Euclidean distance

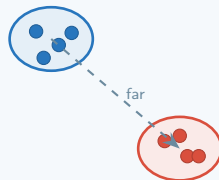
$$\text{dist}(\vec{x}, \vec{y}) = \|\vec{x} - \vec{y}\|_2^2$$

A Valid Distance Metric Must Satisfy

- 1 **Symmetry:** $d(x, y) = d(y, x)$
- 2 **Non-negativity:** $d(x, y) \geq 0$
- 3 **Identity:** $d(x, y) = 0 \Leftrightarrow x = y$
- 4 **Triangle ineq.:**
 $d(x, z) + d(z, y) \geq d(x, y)$

Key Insight

Clustering results are **crucially dependent** on the chosen similarity measure.



Types of Clustering Algorithms

Partition Algorithms (Flat)

- **K-Means** — iterative centroid-based
- Mixture of Gaussians
- Spectral Clustering

All points assigned to exactly one of K clusters simultaneously.

Hierarchical Algorithms

- **Bottom-up** (agglomerative)
Start with N singletons; merge closest pairs
- **Top-down** (divisive)
Start with one cluster; split recursively

Produces a dendrogram of cluster relationships.

K-Means Algorithm

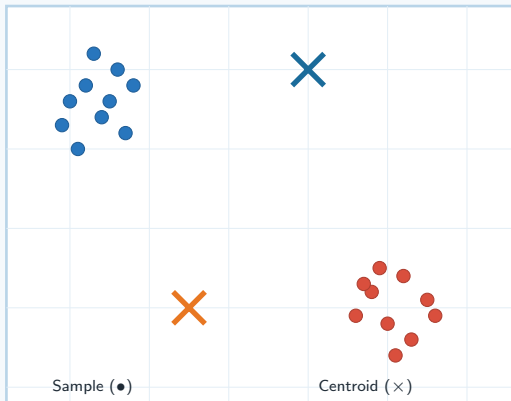
An iterative algorithm that alternates between two simple steps.

- ➊ **Initialize:** Pick K random points as cluster centroids ($\times$)
- ➋ **Alternate:**
 - (a) **Assign** — each data point ($\bullet$) goes to its nearest centroid
 - (b) **Update** — move each centroid ($\times$) to the mean of its points
- ➌ **Stop** when no point changes its cluster assignment

Objective Minimised

$$J = \sum_{i=1}^K \sum_{\vec{x} \in C_i} \|\vec{x} - \vec{\mu}_i\|^2 \qquad \vec{\mu}_i = \frac{1}{|C_i|} \sum_{\vec{x} \in C_i} \vec{x}$$

K-Means: Step 1 — Initialise ($K = 2$)



Legend

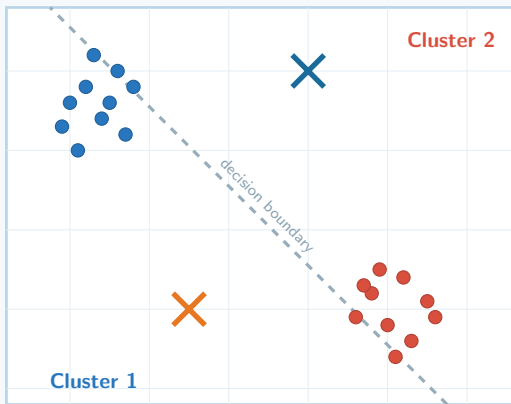
• = data sample

× = centroid

$K = 2$ centroids are placed *randomly* — they may start far from any real cluster centre.

The algorithm will move them iteratively to better positions.

K-Means: Step 2a — Assign Samples to Nearest Centroid

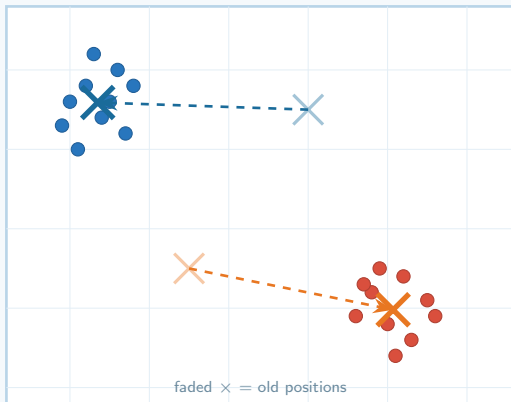


Each sample $\bullet$ is coloured by its nearest centroid $\times$.

$$d(\bullet, \times_1) \leq d(\bullet, \times_2)$$

The dashed line is the **decision boundary** — equidistant from both centroids.

K-Means: Step 2b — Update Centroid Positions

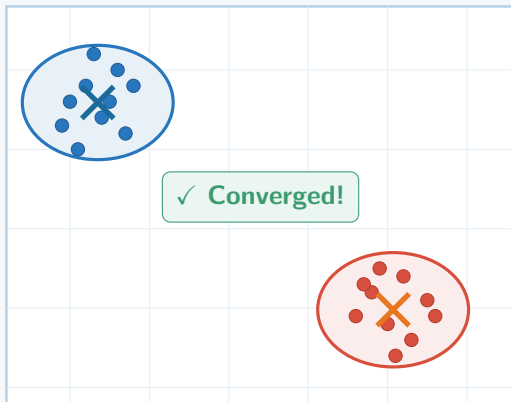


Each centroid $\times$ moves to the **mean** of its assigned samples $\bullet$.

$$\vec{\mu}'_i = \frac{1}{|C_i|} \sum_{\vec{x} \in C_i} \vec{x}$$

Faded crosses show the old (bad) positions. Arrows show the movement direction.

K-Means: Convergence



No sample changed its cluster — the algorithm has **converged**.

The shaded halos highlight the two final clusters.

Centroids $\times$ are now at the **true centres** of the point clouds.

Properties of K-Means

Running Time per Iteration

- 1 Assign samples to nearest centroid: $O(KN)$
- 2 Update centroid positions: $O(N)$

Total per iteration: $O(KN)$

Overall (I iterations): $O(KNI)$

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Local Optimum Only

K-means does **not** guarantee the **global** optimum — it converges to a local minimum of J .

Initialization Pitfalls

- Poor start $\Rightarrow$ poor local optimum
- Risk of empty clusters
- Unbalanced cluster sizes

Challenges and Considerations

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Fundamental Limitations

- Must specify K in advance
- Assumes **spherical**, convex clusters
- Fails on non-convex shapes

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Better Initialization Strategies

- Multiple random restarts
- **K-means++** — spread initial centroids
- Variance-based split / merge

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When K-Means Fails

Concentric rings or crescent shaped clusters — Euclidean distance cannot separate them correctly.

Choosing K — The Elbow Method

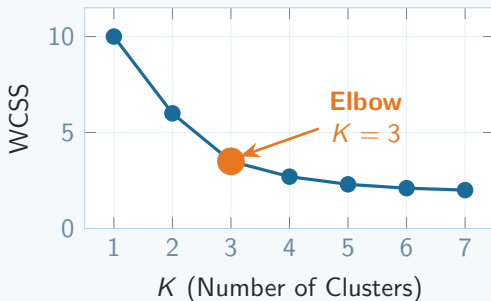
Key Idea: Run K-means for $K = 1, 2, \dots$ and record:

$$J(K) = \sum_{i=1}^K \sum_{\vec{x} \in C_i} \|\vec{x} - \vec{\mu}_i\|^2$$

The “**elbow**” is where J stops dropping sharply — that K is optimal.

Steps:

- 1 Run K-means for each K
- 2 Record WCSS
- 3 Plot K vs. WCSS
- 4 Pick the **elbow**



Note

The elbow can sometimes be ambiguous — always combine with domain knowledge.

Applications of K-Means

Image Segmentation

- Partition image into regions
- Colour quantization
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Gene Expression Analysis

- Identify gene patterns
- Disease classification
- Drug discovery research

Summary

- Clustering is an **unsupervised** learning technique — no labels needed
- K-means is simple, efficient, and widely used in practice
- Key considerations:
 - ▷ Choice of K — use the **Elbow Method**
 - ▷ Distance metric selection
 - ▷ Initialisation strategy (prefer K-means++)
- Hierarchical clustering offers a label-free alternative
- Different algorithms suit different data structures

Key Takeaways

- 1 K-means **minimises** within-cluster sum of squares (WCSS)
- 2 Convergence is **guaranteed**, but only to a **local** optimum
- 3 Complexity: $O(KNI)$ — K clusters, N points, I iterations
- 4 The choice of distance metric is **crucial**
- 5 Use the **Elbow Method** to select K
- 6 Always run **multiple initialisations** (or use K-means++)

Elbow Method — Quick Recap

Plot K vs. WCSS and pick the K where the curve stops dropping sharply (the “elbow”). Diminishing returns beyond that point suggest extra clusters are not meaningful.

Thank You!

Questions?

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